

STRUCTURAL EXHAUSTION: AUDITABLE LOCAL-TO-GLOBAL ESTIMATES FOR HUMAN-AI PDE PROOFS

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ABSTRACT. We introduce *structural exhaustion*, a human-AI methodology for deriving global analytic conclusions from exhaustive systems of local branches. A branch records its space-time window, quantitative hypotheses, gauge and compactness choices, active defect, target estimate, monotone resources, and supporting evidence. A transition either proves a local estimate or converts its first failure into one of finitely many structured defects. Each defect is then closed by a local estimate, a capacity comparison, a decrease of a well-founded resource, or a rigidity contradiction. Exhaustion of all defect branches yields the global estimate; it is not assumed as an external input.

The formalism separates logical coverage, quantitative propagation, and termination. A finite closure theorem transports local bounds to the root. A first-failure compactness theorem shows that if every normalized counterexample sequence enters a terminal contradiction branch, then a uniform estimate holds. A resource theorem makes exhaustion finite when every nonclosing route decreases a discrete rank or consumes a fixed amount of a bounded quantity. Branch-preserving repair remains available: if a step asserting P fails, its descendants survive on the P -branch while $\neg P$ becomes a new structured route.

We give a finite-energy Stokes example in which standard local solenoidal estimates are assembled with a pressure decomposition and an explicit audit of the only entire divergence-curl kernel. Finite energy and an expanding-ball Caccioppoli estimate exclude that kernel, after which the equation closes the harmonic pressure tail. Three companion Navier-Stokes proof graphs then serve as a case study in routing concentration, scale and translation escape, profile splitting, weak-to-strong convergence defects, and pressure-gauge defects to local remedies or resource-decreasing successors. They illustrate the methodology and do not assert a Navier-Stokes regularity theorem.

1. INTRODUCTION

Long proofs in nonlinear PDE are difficult to develop with language models for a reason that is only partly linguistic. Their conclusions depend on choices that remain implicit in ordinary prose: the cylinder on which an estimate is valid, the scale and frame, the pressure gauge, the quotient representative, the topology used at a compactness limit, and the alternative left after an estimate fails. A locally plausible paragraph can silently change one of these choices and thereby change the theorem.

Current AI systems for mathematical reasoning emphasize formal proof search, autoformalization, retrieval, and repair of rejected proof scripts [WJL⁺22, JWZ⁺22, FRRB23, LDZ24]. Recent agentic systems recursively decompose informal arguments into formal subgoals or interleave informal exploration with Lean verification [VVS⁺25, OFS⁺25, ABD⁺25]. Structural exhaustion addresses an earlier analytic decomposition: before a PDE subgoal can be sent to a verifier, its window, topology, gauge, failure mechanism, and quantitative resource must be fixed. Recent PDE-facing systems instead translate control requests or generate numerical solvers [SSX⁺25, LMS⁺25]. These are important but different problems. The present paper addresses the intermediate activity in which a mathematician and an AI system jointly construct a rigorous analytic argument whose decisive estimates may remain in conventional mathematical language.

2020 *Mathematics Subject Classification.* Primary 68T01; Secondary 35Q30, 35B65, 68V15.

Key words and phrases. AI-assisted mathematics, structural exhaustion, local-to-global estimates, loss of compactness, partial differential equations, Stokes equations.

Structural exhaustion represents the proof as an exhaustive directed graph of local analytic states. Nodes are not fragments of prose. Each node exposes the window, normalization, topology, quantitative budget, and active defect required to interpret its claim. Each edge either propagates an estimate or extracts structured information from its first failure. A terminal supplies a local bound or contradiction. When every failure route terminates and the constants or charges are controlled along the graph, the terminal conclusions assemble into a global estimate. The representation is especially natural for concentration compactness, profile decompositions, local pressure splittings, scale selection, and rigidity arguments, where the analysis already advances by alternatives [Lio84a, Lio84b, Gér98, BG99, Ker01].

The constructive step is to use failure as data. If compactness in a topology X is needed downstream, its failure is not recorded as a generic gap. It must yield an exhaustive alternative such as concentration, escape of scale or center, profile splitting, a nonzero defect measure, or loss of a pressure representative. Each alternative carries its inherited bounds and is routed to a local estimate, a new normalized extraction, a finite-capacity account, or rigidity. Thus a missing monolithic estimate can be replaced by local inequalities whose failure modes are structurally exhausted.

The same representation gives a stable repair rule. Suppose a node was intended to establish properties X and Y , and every later step uses only the declared interface $X \wedge Y$. Discovering that the attempted proof does not establish Y leaves the conditional continuation intact. The preceding case is replaced by the exhaustive alternatives

$$\neg X, \quad X \wedge \neg Y, \quad X \wedge Y.$$

The existing continuation is attached unchanged to the third alternative. The first two alternatives are new analytic routes. Their closure may use precisely the same first-failure and resource mechanisms that produced the original estimate. We call this operation *branch-preserving repair*.

The paper makes four contributions.

- (1) It defines a PDE-specific structural-exhaustion state carrying local windows, quantitative interfaces, compactness data, and monotone resources.
- (2) It proves semantic soundness, quantitative local-to-global closure, a first-failure compactness principle, resource-controlled termination, their synthesis as an estimate principle, and branch-preserving repair.
- (3) It gives an operational division of labor: AI proposes and maintains local branches, while the mathematician controls theorem statements, admissible imports, and the acceptance of analytic evidence.
- (4) It supplies a finite-energy Stokes instantiation in which standard local velocity estimates, representative control, and harmonic pressure-tail closure are composed in one auditable graph. Three larger Navier–Stokes graphs show how the same method routes distinct losses of compactness through local remedies and resource-decreasing successors.

The method is deliberately lighter than formalization. Every local inequality, compactness extraction, and rigidity step still requires mathematical proof. Its new role is compositional: it supplies a disciplined way to construct a global estimate from those local ingredients when no suitable global theorem is available. Graph consistency alone is not evidence, but exhaustive, quantitative closure is a proof.

2. STRUCTURAL-EXHAUSTION STATES

2.1. Semantic setting. Fix a universe $\mathcal{U}$ of admissible mathematical objects. An element may be a weak solution, a rescaled sequence, a limiting profile, or a local space–time configuration. A theorem has the form

$$A(s) \implies T(s), \quad s \in \mathcal{U}, \tag{2.1}$$

where A is the theorem hypothesis and T its conclusion. Direct and contradiction arguments are both included: in the latter case, the active goal is the emptiness of $A \wedge \neg T$.

Definition 2.1 (PDE branch state). A branch state is a record

$$B = (W, H, \Gamma, \tau, O, Q, M, E).$$

Here W fixes the local window, scale, and coordinate frame; H is the list of analytic hypotheses already available on that window; Γ records gauge, quotient, normalization, and convergence choices; τ is the active case predicate; and O is the qualitative goal on this case. The quantitative interface $Q = (q_B, d_B)$ consists of finite nonnegative functions on $\llbracket B \rrbracket$, recording a target quantity and its admissible input datum. The resource record M contains the energy, mass, rank, scale count, or other monotone quantity used to control repeated failure. Finally, E contains the mathematical evidence accepted for the preceding implications. The semantic region is

$$\llbracket B \rrbracket = \{s \in \mathcal{U} : W(s) \wedge H(s) \wedge \Gamma(s) \wedge \tau(s)\}.$$

The fields are mathematical data. They may be written in prose, equations, or a table; no particular serialization is part of the method.

The separation between H , Γ , τ , Q , and M matters in PDE. A bound can survive rescaling while a gauge choice does not; a sequence can converge weakly in the energy space without giving the strong convergence needed for a nonlinear term; and a pressure statement can be meaningful only modulo functions of time. A defect extraction may preserve the analytic hypotheses but consume critical mass or lower a scale rank. Recording these choices prevents a downstream argument from consuming a stronger statement or more quantitative resource than its parent produced.

Definition 2.2 (Valid transition). A transition from a branch B to children $B_1, \dots, B_m$, written

$$B \Longrightarrow B_1 \vee \dots \vee B_m,$$

is valid if its evidence establishes

$$\llbracket B_i \rrbracket \subseteq \llbracket B \rrbracket \quad (1 \leq i \leq m), \quad \llbracket B \rrbracket \subseteq \bigcup_{i=1}^m \llbracket B_i \rrbracket. \quad (2.2)$$

The first condition is refinement and the second is exhaustiveness. Children may overlap, although disjoint alternatives are preferable for diagnosis. A one-child transition is an implication; a multi-child transition is a case split.

Definition 2.3 (Quantitative transition). Let $B \Longrightarrow B_1 \vee \dots \vee B_m$ be valid. It is a *routing estimate* if its evidence supplies nonnegative constants $a_B, b_{B_i}, \kappa_{B_i}$ such that, for $s \in \llbracket B \rrbracket$,

$$q_B(s) \leq a_B d_B(s) + \max_{i: s \in \llbracket B_i \rrbracket} b_{B_i} q_{B_i}(s), \quad d_{B_i}(s) \leq \kappa_{B_i} d_B(s). \quad (2.3)$$

It is an *aggregate estimate* if

$$q_B(s) \leq a_B d_B(s) + \sum_{i=1}^m b_{B_i} q_{B_i}(s), \quad \sum_{i=1}^m d_{B_i}(s) \leq \kappa_B d_B(s). \quad (2.4)$$

In an aggregate estimate, q_{B_i} and d_{B_i} are extended by zero outside $\llbracket B_i \rrbracket$. The second inequality in (2.4) includes the overlap or multiplicity bound needed to sum local estimates. A transition may carry both interfaces when it first chooses a case and then decomposes its local pieces.

Every transition also declares an *output interface*: the facts that its children may use. A proof below the transition is interface-respecting when it uses the transition only through those facts, not through an unstated detail of the argument that produced them. This familiar modularity condition is what makes branch-preserving repair possible.

Definition 2.4 (Structurally exhausted proof graph). A finite rooted directed acyclic graph is structurally exhausted for (2.1) if:

- (1) $\{s : A(s)\} \subseteq \llbracket B_0 \rrbracket$ for its root B_0 ;
- (2) every nonterminal node has a valid transition to all of its children;
- (3) every terminal L has accepted evidence for $\llbracket L \rrbracket \subseteq \{s : T(s)\}$.

A terminal may close by a direct estimate, a capacity contradiction, a rigidity theorem, or identification with an explicitly classified exceptional case. For a quantitative conclusion it also records a bound $q_L \leq c_L d_L$, or proves that $\llbracket L \rrbracket$ contains no normalized failure.

The graph is both a proof presentation and a composition rule for estimates. Logical coverage, local analytic evidence, resource consumption, and constant propagation remain separately reviewable.

2.2. PDE transitions. The states used in this paper admit five recurring transitions.

- (1) A *localization* replaces a global object by nested cylinders or frequency windows and records every cutoff error.
- (2) A *decomposition* splits velocity, pressure, profiles, scales, or defects into named components and retains the remainder.
- (3) An *estimate* proves a local inequality and records its constant, data dependence, and overlap multiplicity.
- (4) A *compactness or rigidity step* changes the state only in the topology actually obtained and records the property transported to the limit.
- (5) A *decision* introduces exhaustive positive and negative cases for a property not yet known; its negative case identifies a defect or a resource-decreasing successor.

For each node a reader should be able to answer eight questions: what case is active, what inputs are available, which target quantity is controlled, which resource is retained or consumed, what new fact is proved, which result proves it, which alternatives leave the node, and where each alternative closes. In the Stokes example these questions are answered by Table 3; the larger Navier–Stokes instances use the same format.

2.3. Human–AI division of labor.

Protocol 2.5 (Structural-exhaustion loop). Starting from (2.1), the collaborators repeat the following steps.

- (1) The mathematician fixes the theorem, admissible solution class, and which external results may be cited.
- (2) The AI proposes the next local estimate or exhaustive split, together with its input/output interface, constant dependence, resource change, and a source-level argument.
- (3) The mathematician accepts, rejects, or weakens that argument. Simple graph checks may verify numbering, acyclicity, coverage declarations, and source links, but acceptance of an analytic estimate remains mathematical review.
- (4) An accepted transition is added. A rejected assertion P is replaced by the split $P \vee \neg P$; the existing continuation is retained on P , while $\neg P$ is converted into a defect state whose local data and resource record determine the next route.
- (5) The process stops only when every reachable branch has a terminal proof of the target.

This allocation uses the model where it is effective: proposing decompositions, searching for missing cases, restating local obligations, tracing citations, and maintaining a large graph. It leaves theorem ownership and analytic judgment with the human author. In particular, a fluent proof sketch is never itself evidence that a branch is closed.

3. LOCAL-TO-GLOBAL CORRECTNESS AND REPAIR

Theorem 3.1 (Soundness). *If a finite graph is structurally exhausted for (2.1), then $A(s) \Rightarrow T(s)$ for every $s \in \mathcal{U}$.*

Proof. Fix s with $A(s)$. The root condition gives $s \in \llbracket B_0 \rrbracket$. At every nonterminal node containing s , exhaustiveness in (2.2) gives a child containing s . Finiteness and acyclicity produce a terminal L with $s \in \llbracket L \rrbracket$. Terminal closure then gives $T(s)$. $\square$

Theorem 3.2 (Quantitative local-to-global closure). *Let a finite structurally exhausted graph carry quantitative interfaces $Q_B = (q_B, d_B)$. Suppose every nonterminal transition satisfies either (2.3) or (2.4), and every terminal L satisfies*

$$q_L(s) \leq c_L d_L(s), \quad s \in \llbracket L \rrbracket.$$

Then the root satisfies a uniform estimate

$$q_{B_0}(s) \leq C_{\text{root}} d_{B_0}(s), \quad s \in \llbracket B_0 \rrbracket, \quad (3.1)$$

where C_{root} is determined by the terminal constants and the recorded transition constants. In particular, the global constant is an output of the closed local graph.

Proof. Order the vertices backwards from the terminals. Put $C_L = c_L$ at a terminal. At a routing transition define

$$C_B = a_B + \max_i b_{Bi} \kappa_{Bi} C_{B_i},$$

and at an aggregate transition define

$$C_B = a_B + \kappa_B \max_i b_{Bi} C_{B_i}.$$

The routing formula follows by choosing a child containing s , using its bound, and transporting its datum through (2.3). For an aggregate transition,

$$\sum_i b_{Bi} q_{B_i} \leq \max_i (b_{Bi} C_{B_i}) \sum_i d_{B_i} \leq \kappa_B \max_i (b_{Bi} C_{B_i}) d_B.$$

Backward induction is valid because the graph is finite and acyclic, and gives (3.1). $\square$

The preceding theorem assembles inequalities already visible on the graph. A compactness argument often creates those inequalities indirectly by excluding all normalized failures.

Theorem 3.3 (First-failure compactness closure). *Let $\mathcal{U}$ be a cone and let $q, d : \mathcal{U} \rightarrow [0, \infty)$ be positively homogeneous of the same degree $\alpha > 0$. Assume that every sequence satisfying*

$$q(s_n) = 1, \quad d(s_n) \longrightarrow 0, \quad (3.2)$$

admits a subsequence that enters a finite exhaustive extraction graph. Each edge of this graph must preserve the normalization and the topology, gauge, and resource data required by its successor. If every terminal contradicts (3.2), then there is a finite constant C such that

$$q(s) \leq C d(s), \quad s \in \mathcal{U}. \quad (3.3)$$

Proof. If (3.3) failed, there would be u_n with $q(u_n) > n d(u_n)$. Setting $s_n = q(u_n)^{-1/\alpha} u_n$ gives $q(s_n) = 1$ and $d(s_n) < 1/n$. Exhaustiveness routes a subsequence to a terminal, while terminal closure says that no such normalized sequence can reach that terminal. This contradiction proves the estimate. $\square$

Theorem 3.4 (Resource-controlled exhaustion). *Consider a finitely branching structural-exhaustion tree. Suppose every live state carries a rank $r \in \{0, \dots, R\}$ and a budget $0 \leq \mu \leq M$. Along every nonclosing edge either r decreases, or r is unchanged and μ decreases by at least $\delta > 0$. When r decreases, μ may be reset to at most M . Then every path has length at most*

$$(R + 1)(1 + \lfloor M/\delta \rfloor),$$

and the tree is finite. Consequently, if every terminal supplies a local estimate or contradiction, structural exhaustion produces an unconditional global conclusion.

Proof. At each fixed rank, at most $\lfloor M/\delta \rfloor$ budget-consuming edges are possible. The rank can decrease at most R times. The displayed bound follows. A finitely branching tree with uniformly bounded path length has finitely many vertices. $\square$

Theorem 3.5 (Structural-exhaustion estimate principle). *Let $\mathcal{U}$ be a cone and let $q, d : \mathcal{U} \rightarrow [0, \infty)$ be positively homogeneous of the same positive degree. Suppose that every normalized failure sequence*

$$q(s_n) = 1, \quad d(s_n) \rightarrow 0,$$

enters a finitely branching extraction tree whose transitions preserve the normalization and every topology, gauge, and quantitative interface consumed downstream. Assume further that:

- (1) *every nonclosing transition either lowers a rank $r \in \{0, \dots, R\}$ or, at fixed rank, consumes at least $\delta > 0$ from a budget bounded by M ; and*
- (2) *every terminal contradicts the normalized failure conditions.*

Then the extraction tree is finite and there is a constant $C < \infty$ such that

$$q(s) \leq Cd(s), \quad s \in \mathcal{U}.$$

If this estimate is the terminal interface of a finite quantitative proof graph, Theorem 3.2 propagates it, together with all recorded overlap and transport constants, to the root.

Proof. The resource hypotheses and finite branching imply finiteness of the extraction tree by Theorem 3.4. It is therefore a finite exhaustive extraction graph of the kind required in Theorem 3.3. Terminal exclusion gives the asserted estimate. The last statement is exactly Theorem 3.2. $\square$

The same routing argument gives a sharp residual cover for an incomplete proof.

Theorem 3.6 (Partial closure). *Suppose every transition in a finite rooted graph is valid, but a set $\mathcal{L}_{\text{open}}$ of reachable leaves has not yet been closed. If all other leaves prove T , then*

$$\{s : A(s) \wedge \neg T(s)\} \subseteq \bigcup_{L \in \mathcal{L}_{\text{open}}} \llbracket L \rrbracket. \quad (3.4)$$

Thus the open leaves, rather than an informal list of gaps, cover every residual class in which a counterexample may remain. Equality holds when the root is exactly A and the branch alternatives form a disjoint partition.

Proof. Route a putative s with $A(s) \wedge \neg T(s)$ from the root as in the proof of Theorem 3.1. It cannot reach a closed leaf, so it reaches an open one. $\square$

3.1. Branch-preserving repair. Dependency-graph language often suggests that a failed lemma invalidates its forward dependency closure. That is not the repair rule here. The downstream proofs are conditional arguments, and a failed attempt to establish their input does not make those conditional arguments false.

Theorem 3.7 (Branch-preserving repair). *Let B be the last verified branch before a transition intended to establish a predicate P . Assume every descendant of that transition is interface-respecting: it uses the transition only through P and the facts already present in B . If the evidence for P fails, replace the transition by*

$$B \implies B^+ \vee B^-, \quad \llbracket B^+ \rrbracket = \llbracket B \rrbracket \cap \{P\}, \quad \llbracket B^- \rrbracket = \llbracket B \rrbracket \cap \{\neg P\}. \quad (3.5)$$

Then:

- (1) *the replacement transition is valid independently of the failed evidence;*
- (2) *the entire former descendant graph remains valid, without alteration, below B^+ ;*

- (3) the only new proof obligation is B^- ; and
- (4) if the former descendant graph closed B^+ , the repaired graph satisfies (3.4) with B^- as its additional residual class.

Proof. The two sets in (3.5) refine $\llbracket B \rrbracket$ and cover it by excluded middle. On B^+ , the predicate P is part of the case definition. Every former descendant therefore receives exactly the interface it previously consumed, so its proof is unchanged. No claim has been made on B^- ; it is the new open leaf. The final assertion is [Theorem 3.6](#). $\square$

[Figure 1](#) shows the repaired local graph. The descendant subgraph is not regenerated; it is reattached to the case in which its declared input P holds.

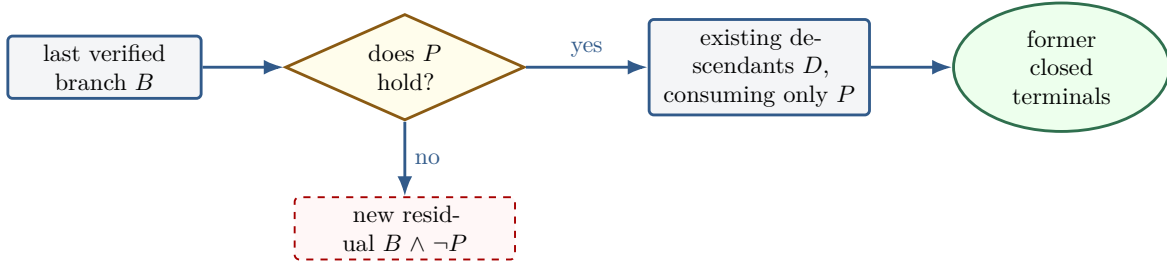


FIGURE 1. Branch-preserving repair. The positive arm retains every interface-respecting descendant; the negative arm is the only new analytic obligation.

Corollary 3.8 (Ordered first-failure repair). *If the failed node asserted $P = X_1 \wedge \dots \wedge X_m$, it may be replaced by the disjoint exhaustive split*

$$\begin{aligned}
 &\neg X_1, \\
 &X_1 \wedge \neg X_2, \\
 &\vdots \\
 &X_1 \wedge \dots \wedge X_{m-1} \wedge \neg X_m, \\
 &X_1 \wedge \dots \wedge X_m.
 \end{aligned} \tag{3.6}$$

The former descendants remain on the final branch. Each new branch records the first absent property and can be analyzed independently.

For the motivating two-property example, discovering that Y is not established creates the case $X \wedge \neg Y$; it does not alter any theorem proved under $X \wedge Y$. The additional case $\neg X$ is retained when X was also part of the unproved conjunctive node. If X was already verified earlier, that branch is empty and may be closed immediately.

Theorem 3.9 (Monotone closure). *Assume repairs are made by [Theorem 3.7](#) and accepted evidence is changed only when its own statement or proof is found defective. Closing or refining one open branch does not reopen any other closed branch. If every newly created complementary branch is eventually closed, the resulting graph proves the original theorem.*

Proof. A repair partitions one semantic region and leaves all other regions and their evidence unchanged. Refinement preserves this separation. When all leaves are closed, [Theorem 3.1](#) applies. $\square$

The role of dependency information is therefore diagnostic rather than destructive. It identifies the interface at which to insert the dichotomy and the continuation to attach to its positive arm. It does not mark a forward set of mathematical statements as invalid.

4. SPECIALIZATION TO PDE PROOFS

For a PDE argument, the abstract record in [Definition 2.1](#) is made concrete as follows.

TABLE 1. Minimal information carried by a PDE branch.

Field	Typical content	Question answered
W	Cylinder, cutoff, scale, frame, time orientation	Where is the claim valid?
H	Equation, solution class, bounds, sign or flux assumptions	Which analytic hypotheses are available?
Γ	Pressure gauge, quotient, normalization, convergence topology	Which representative and limiting meaning are fixed?
τ	Profile type, scale relation, active defect, positive/negative case	Which alternative is being analyzed?
O	Regularity, decay, contradiction, classification, or reduction	What must this branch prove?
Q	Target seminorm, input norm, constants, and overlap multiplicity	Which quantitative estimate is propagated?
M	Energy, critical mass, profile count, scale rank, or failure budget	Why must repeated refinement terminate?
E	Lemma, calculation, citation, and exact input/output interface	Why may the reader traverse the incoming edge?

Three constraints are particularly important.

Locality. An interior estimate consumes data on an explicitly larger window and produces a conclusion on an explicitly smaller one. Cutoff errors and pressure remainders are children of the same node unless a cited estimate controls them.

Representative stability. Statements involving pressure, harmonic gradients, potentials, or quotient spaces specify the representative that survives the transition. Equality of vorticity does not, by itself, identify velocity; weak convergence of pressure does not choose its time-dependent additive constant.

Topological sufficiency. Each limit passage names the convergence needed by the next nonlinear operation. A missing strong convergence statement is handled as a negative branch, not silently upgraded from a weak one.

These constraints turn common review questions into branch questions. If a profile decomposition omits a scale relation, split on that relation. If a pressure estimate controls only the nonharmonic component, retain the harmonic component as a branch. If a rigidity lemma requires $X \wedge Y$, use the ordered alternatives in (3.6). In each case the successful continuation remains available while the complementary analytic class becomes the next research target.

This is the connection between structural exhaustion and loss of compactness. A compactness step declares the limit object, its normalization, the topology of convergence, the properties transported to the limit, and the resource retained after extraction. If one part of this interface fails, the first failure is converted into positive structure: concentration, escape of scale or center, profile splitting, a nonzero defect measure, or loss of a pressure representative. The successful compactness continuation remains available, while each defect is routed to a new local estimate, normalization, capacity comparison, or rigidity test.

The method therefore replaces a missing global compactness estimate by an exhaustive local construction. A concentration branch consumes a fixed amount of critical mass; an escaping branch changes frame or becomes locally invisible; a splitting branch produces finitely many active profiles;

a weak-to-strong defect produces another core; and a gauge defect is separated from the velocity limit and closed elliptically. If these routes exhaust the normalized failures, [Theorem 3.3](#) yields the estimate. If every continuing route also decreases a rank or consumes a bounded resource, [Theorem 3.4](#) makes the exhaustion finite. The methodology is thus the local-to-global tool: it constructs the global conclusion from local analytic mechanisms rather than presupposing a theorem that already contains it.

5. COMPLETE STOKES INSTANTIATION

The example now isolates the nonlocal pressure issue that occurs in the Navier–Stokes developments while removing the nonlinear term. On $\mathbb{R}^3 \times (0, T)$ consider

$$\partial_t u - \Delta u + \nabla p = -\operatorname{div} F, \quad \operatorname{div} u = 0, \quad (5.1)$$

where $(\operatorname{div} F)_i = \partial_j F_{ij}$. The pressure is defined modulo a distribution depending only on time.

Definition 5.1 (Finite-energy Stokes state). A finite-energy weak solution of (5.1) satisfies

$$u \in L_{\operatorname{loc}}^\infty(0, T; L^2(\mathbb{R}^3)) \cap L_{\operatorname{loc}}^2(0, T; \dot{H}^1(\mathbb{R}^3)), \quad p \in \mathcal{D}'(\mathbb{R}^3 \times (0, T)),$$

and the equation holds in distributions. We assume $F \in C^\infty(\mathbb{R}^3 \times (0, T))$, with its derivatives locally square integrable in time and space. This is the linear analogue of the finite-energy class used in the Navier–Stokes applications.

Fix $B_r \Subset B_\rho \Subset B_R$, choose $\chi \in C_c^\infty(B_R)$ with $\chi = 1$ on B_ρ , and let $N(x) = (4\pi|x|)^{-1}$, so $-\Delta N = \delta_0$. Define the compact-source pressure and the pressure tail by

$$p_{\text{near}} = \partial_i \partial_j N * (\chi F_{ij}), \quad p_{\text{tail}} = p - p_{\text{near}} - c(t), \quad (5.2)$$

where the time distribution c fixes the gauge, for example by requiring $\langle p_{\text{tail}}(\cdot, t), \varphi \rangle_x = 0$ for a fixed $\varphi \in C_c^\infty(B_\rho)$ with $\int \varphi = 1$. Then $\Delta p_{\text{tail}} = 0$ on B_ρ . The tail contains all influence not represented by the compactly supported local source; its regularity will be proved, not assumed.

Remark 5.2 (Why finite energy is essential). For a nonsmooth scalar function $a(t)$,

$$u(x, t) = a(t)e_1, \quad p(x, t) = -a'(t)x_1, \quad F = 0$$

is a local distributional Stokes solution with zero vorticity. It is excluded from [Definition 5.1](#) unless $a = 0$, because a nonzero spatially constant velocity is not in $L^2(\mathbb{R}^3)$. Thus a local curl argument alone cannot prove the theorem below; the finite-energy branch must be used.

For nested cylinders $Q_r \Subset Q_R$ and an integer $m \geq 0$, set

$$q_m(Q_r; u) := \sum_{2j+|\alpha| \leq m} \|\partial_t^j \partial_x^\alpha u\|_{L^2(Q_r)}, \quad (5.3)$$

$$d_m(Q_R; u, F) := \|u\|_{L^2(J_R; L^2(\mathbb{R}^3))} + \mathcal{N}_{m, R, J_R}(F). \quad (5.4)$$

Here $\mathcal{N}_{m, R, J_R}(F)$ denotes a fixed finite sum of local L^2 -norms of the derivatives of F required up to parabolic order m . These are the quantitative interface of the Stokes graph. Its resource record is the finite derivative rank together with the remaining cylinder buffer. [Lemma 5.7](#) supplies the local transition $q_m(Q_r; u) \leq C_m d_m(Q_R; u, F)$, including its buffer and data dependence. The whole-space finite-energy hypothesis has a separate role: it excludes every entire divergence–curl ambiguity by [Lemma 5.6](#). The example therefore does not infer a global bound for normalized difference quotients from the energy of u ; it composes a proved local estimate with a proved global representative exclusion.

Theorem 5.3 (Finite-energy Stokes regularity by local pressure-tail closure). *Let (u, p, F) be a finite-energy Stokes state on $\mathbb{R}^3 \times (0, T)$. Then*

$$u \in C^\infty(\mathbb{R}^3 \times (0, T)), \quad \nabla p \in C^\infty(\mathbb{R}^3 \times (0, T)).$$

For every compact cylinder $Q' \Subset \mathbb{R}^3 \times (0, T)$ and every derivative order, the corresponding norms of u and ∇p are bounded by the finite-energy norm of u on a larger time interval and finitely many local norms of derivatives of F on a buffered spatial cylinder. For every split (5.2), $\nabla p_{\text{tail}} \in C^\infty(B_r \times I_r)$.

The proof uses no whole-space pressure formula and no global heat or Oseen kernel. Compactly supported divergence corrections remove pressure from the velocity estimates and record their constants on nested cylinders. The finite-energy kernel lemma separately excludes an unrecorded entire harmonic representative. With these two obligations closed, the equation controls the harmonic pressure tail on the target cylinder.

5.1. How to read the proof architecture. Node [2] records the local near/tail pressure split without assigning any time regularity to the harmonic tail. Curl and local heat estimates give the smooth vorticity in [3]. Solenoidal cutoffs and annular Bogovskiĭ corrections give the quantitative local bootstrap in [4], hence smoothness of the raw velocity in [5]. Independently, [6] excludes the entire finite-energy divergence–curl kernel, so [7] certifies that no global harmonic representative was lost. Node [8] then uses the original equation to prove smoothness of the pressure tail, and [9] gives full regularity of the unmodified velocity and pressure gradient.

5.2. Diagram map. Table 2 records the single diagram and its node range.

TABLE 2. Map of the Stokes proof-flow diagram.

Diagram	Nodes	Role
Figure 2	[1]–[9]	Split the pressure locally, propagate the solenoidal velocity estimate, exclude the entire finite-energy harmonic kernel, and recover the pressure tail from the equation.

5.3. Proof-dependency diagram. Figure 2 displays the local analytic spine and the separate representative audit used by the final conclusion.

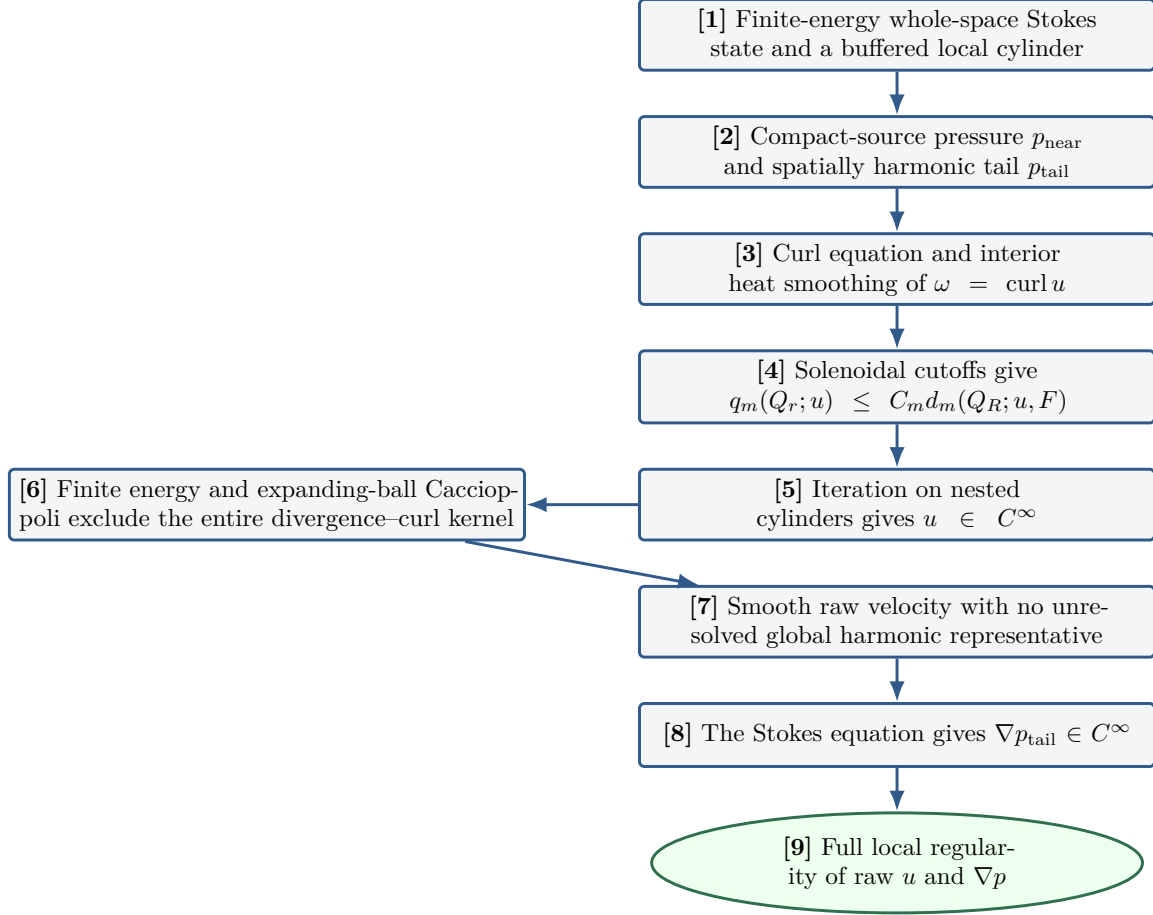


FIGURE 2. Local closure of the Stokes pressure tail. Quantitative velocity estimates and global representative control are verified separately before the original equation recovers the unmodified pressure gradient.

5.4. **Node-by-node audit table.** Table 3 identifies the precise lemma supporting every state, transition, and terminal closure.

Table 3: Source ledger for Figure 2.

Node	Input	Output	Mathematical source
[1]	Definition 5.1 and a target cylinder	A finite-energy state with fixed local buffers	Hypotheses of Theorem 5.3.
[2]	[1] and a spatial cutoff	$p = p_{\text{near}} + p_{\text{tail}} + c(t)$, with $\Delta p_{\text{tail}} = 0$ locally	Lemma 5.4.
[3]	[1]	Smooth local vorticity	Lemma 5.5.
[4]	[1], [3], nested cylinders, and q_m, d_m	$q_m(Q_r; u) \leq C_m d_m(Q_R; u, F)$, with recorded buffer and data dependence	Lemma 5.7.
[5]	[4] at every finite derivative rank	Smooth raw velocity on the target cylinder	Iteration in Lemma 5.7.

Node	Input	Output	Mathematical source
[6]	[1], [5], and any entire L^2 field with zero divergence and curl	The field vanishes	Lemma 5.6.
[7]	[5], [6]	Smooth raw velocity with no unresolved entire finite-energy harmonic representative	Lemmas 5.6 and 5.7.
[8]	[2], [7]	Smooth harmonic pressure tail	Lemma 5.8.
[9]	[2], [7], [8]	Smooth raw velocity and pressure gradient	Proof of Theorem 5.3.

5.5. Monotone-hypothesis ledger. [Table 4](#) records the data retained while cylinders shrink and the independent kernel audit is closed over expanding balls.

TABLE 4. Hypotheses retained along the Stokes route.

Datum	How it is used	Persistence
Nested cylinders	Support cutoffs, annular divergence corrections, and interior estimates	Windows only shrink.
Finite-energy bound for u	Controls every cutoff estimate and excludes the entire divergence-curl kernel	Retained on every local window and in the whole-space representative audit.
Smooth tensor forcing F	Supplies the local pressure source and the difference-quotient right-hand sides	Differentiated only to finite order.
Quantitative interface q_m, d_m	Records and propagates the cutoff-estimate constant to the target cylinder	Retained at fixed order; the order increases only after the previous estimate closes.
Derivative rank and cylinder buffer	Control iteration and prevent an unrecorded infinite refinement	Finite at each requested order.
Distributional Stokes equation and incompressibility	Remove pressure from solenoidal tests and recover it after velocity closure	Never replaced by a projected whole-space equation.
Pressure modulo functions of time	Makes the conclusion a statement about ∇p	Fixed in (5.2) .

5.6. Structural-exhaustion ledger. [Table 5](#) verifies that every analytic obligation used by the final conclusion is discharged.

TABLE 5. Closure of the analytic obligations in the Stokes graph.

Obligation	Discharged at	Closure
Local velocity control	[4]–[5]	Solenoidal cutoff testing and Bogovskiĭ correction give $q_m \leq C_m d_m$; iteration on nested cylinders makes the raw velocity smooth.
Global representative control	[6]–[7]	Every entire finite-energy divergence-free and curl-free ambiguity vanishes by expanding-ball Caccioppoli.
Pressure-tail closure	[8]–[9]	The original equation makes ∇p_{tail} , and hence ∇p , smooth on the smaller cylinder.

5.7. Analytic proof. All estimates in this subsection are obtained on nested cylinders. The whole-space hypothesis supplies only the finite-energy bound that excludes the parasitic kernel; no whole-space pressure representation or heat kernel is used.

Lemma 5.4 (Localized pressure splitting). *Let $B_r \Subset B_\rho \Subset B_R$, let $\chi \in C_c^\infty(B_R)$ equal one on B_ρ , and define p_{near} and p_{tail} by (5.2). Then*

$$-\Delta p_{\text{near}} = \partial_i \partial_j (\chi F_{ij}) \quad \text{on } \mathbb{R}^3, \quad \Delta p_{\text{tail}} = 0 \quad \text{on } B_\rho.$$

For every spatial derivative order, the local norms of p_{near} on B_ρ are bounded by finitely many local norms of F on B_R .

Proof. Taking divergence in (5.1) gives

$$-\Delta p = \partial_i \partial_j F_{ij}. \quad (5.5)$$

Since $-\Delta N = \delta_0$, the compact-source potential in (5.2) satisfies the first displayed identity. Because $\chi = 1$ on B_ρ , subtraction from (5.5) gives the second. Calderón–Zygmund and interior elliptic estimates give the local bounds [CZ52, Ste70]. The construction uses the Newtonian parametrix only on the compactly supported source χF ; it makes no assertion about a whole-space representative of p . $\square$

Lemma 5.5 (Vorticity equation and local smoothing). *For $\omega = \text{curl } u$,*

$$(\partial_t - \Delta)\omega = -\text{curl div } F \quad (5.6)$$

in distributions. Hence ω is smooth on every compact cylinder in $\mathbb{R}^3 \times (0, T)$.

Proof. Apply curl to (5.1); the pressure term vanishes. Interior hypoellipticity for the heat operator on nested cylinders gives smoothness because F is smooth. This is a local parametrix argument: a cutoff is inserted first, and every commutator is supported away from the target cylinder [LSU68, Chapter IV]. $\square$

Lemma 5.6 (Finite-energy harmonic kernel). *If $H \in L^2(\mathbb{R}^3; \mathbb{R}^3)$ satisfies*

$$\text{div } H = 0, \quad \text{curl } H = 0 \quad \text{in } \mathcal{D}'(\mathbb{R}^3),$$

then $H = 0$.

Proof. The vector identity $-\Delta H = \text{curl curl } H - \nabla \text{div } H$ shows that every component of H is harmonic. Let η_R be one on B_R , supported in B_{2R} , and satisfy $|\nabla \eta_R| \leq C/R$. Testing the harmonic equation with $\eta_R^2 H$ gives the local Caccioppoli estimate

$$\int_{B_R} |\nabla H|^2 \leq \frac{C}{R^2} \int_{B_{2R}} |H|^2 \leq \frac{C}{R^2} \|H\|_{L^2(\mathbb{R}^3)}^2.$$

Letting $R \rightarrow \infty$ yields $\nabla H = 0$. An $L^2(\mathbb{R}^3)$ constant field is zero. $\square$

Lemma 5.7 (Local solenoidal bootstrap). *Let (u, p, F) satisfy Definition 5.1. For nested cylinders $Q_r = B_r \times J_r \Subset B_R \times J_R = Q_R$ and every integer $m \geq 0$, the derivatives of u of parabolic order at most m satisfy an estimate of the form*

$$\sum_{2j+|\alpha| \leq m} \|\partial_t^j \partial_x^\alpha u\|_{L^2(Q_r)} \leq C_{m,r,R,J_r,J_R} \left(\|u\|_{L^2(J_R; L^2(\mathbb{R}^3))} + \mathcal{N}_{m,R,J_R}(F) \right), \quad (5.7)$$

where $\mathcal{N}_{m,R,J_R}(F)$ contains finitely many local L^2 -norms of derivatives of F on Q_R . Consequently $u \in C^\infty(Q_r)$.

Proof. Choose spatial and temporal cutoffs supported in Q_R and equal to one on an intermediate cylinder. Spatial difference quotients of the weak equation cannot be tested directly after multiplication by a cutoff, because the cutoff destroys incompressibility. On the annulus where the cutoff varies, solve

$$\operatorname{div} w = \nabla \eta \cdot D_h u, \quad w|_{\partial A} = 0,$$

with the Bogovskii operator. The corrected test field $\eta D_h u - w$ is compactly supported and divergence-free. Testing the Steklov-averaged equation with this field removes the pressure identically. The local energy estimate, the boundedness of the annular divergence correction, and Young's inequality give the first spatial difference-quotient bound. Repeating the argument for differentiated equations and using (5.6) supplies all spatial derivatives. Temporal difference quotients are then obtained from the same solenoidal weak equation; the spatial estimates absorb the Laplacian term. Iteration over strictly nested cylinders proves (5.7). This is the local cutoff proof of the standard finite-energy Stokes regularity estimate; see [Soh01, Chapters II–III]. In particular, the constants in (5.7) come from the cutoff calculation itself; no compactness normalization of global difference quotients is used. $\square$

Lemma 5.8 (Local closure of the harmonic pressure tail). *On the cylinders of Lemma 5.4,*

$$\nabla p_{\text{tail}} = -\operatorname{div} F - \partial_t u + \Delta u - \nabla p_{\text{near}}.$$

Consequently ∇p_{tail} is smooth on every smaller cylinder, with local estimates inherited from Lemmas 5.4 and 5.7.

Proof. Rearrange the original Stokes equation and subtract ∇p_{near} . Every term on the right is smooth on the smaller cylinder. This determines the gradient of the harmonic tail without constructing the pressure outside the cutoff ball. The additive function $c(t)$ is immaterial. $\square$

Proof of Theorem 5.3. Fix an arbitrary compact cylinder and two buffered cylinders around it. Lemma 5.7 gives smoothness of the original velocity u , not merely of a quotient representative. Independently, if two whole-space finite-energy representatives had the same divergence and curl, their difference would be the field excluded by Lemma 5.6; thus the global representative audit has no residual harmonic branch.

Apply Lemma 5.4 on the buffered spatial balls. The compact-source pressure is smooth. By Lemma 5.8, the harmonic tail has a smooth gradient on the target cylinder. Hence ∇p is smooth there. The target cylinder was arbitrary, which proves the theorem throughout $\mathbb{R}^3 \times (0, T)$. $\square$

6. NAVIER–STOKES COMPACTNESS CASE STUDY

The Stokes graph is small enough to inspect on one page. Three companion Navier–Stokes developments provide a larger case study of how structural exhaustion represents compactness arguments at realistic scale. The graphs record where compactness is available, the mechanisms by which it can fail, the topology obtained at each limit, and the localization, profile extraction, normalization, or rigidity step proposed to repair each failure. They document an LLM-assisted representation of these analytic mechanisms, not a Navier–Stokes regularity theorem or a short substitute for its underlying analytic proofs. Each opens with a proof-first reading guide, an acyclic

diagram, a node-by-node source ledger, a monotone-hypothesis ledger, and a structural-exhaustion table. The counts in Table 6 refer to solid numbered nodes and diagram parts in those source files.

TABLE 6. Navier–Stokes compactness case-study graphs.

Development	Nodes	Diagram parts	Source
Navier–Stokes setup and compactness architecture	39	2	<code>proof_setup.tex</code>
Type I residual-branch argument	159	9	<code>type_I_residual_closure.tex</code>
Type II branch argument	135	12	<code>type_II_regularity.tex</code>

The three developments organize, respectively, the compactness setup, a residual Type I route, and a Type II route. Their common unit is not an open gap but a first-failure interface. The interface specifies the topology needed by the next nonlinear operation; failure of that topology produces one of the structured routes in Table 7. The table is a routing index into the companion developments: it does not assert that the short paper has independently proved the local lemma named in each route.

TABLE 7. Structural exhaustion of compactness defects in the companion graphs.

Failure mechanism	Extracted local data	Structural-exhaustion route	Quantitative role
Concentration	Centered core with a positive critical-mass floor	Retain the core, normalize its frame, and test compactness or rigidity again	Each new active core consumes a fixed amount of finite critical mass.
Translation escape	Center sequence and terminal no-escape data	Recenter the active frame or prove that the escaping component is locally invisible	The target window retains only mass visible in one of the exhaustive frames.
Scale escape or cascade	Ordered inner, comparable, and outer scales	Group comparable scales, pass outer scales to a vanishing route, and promote inner scales to a new core	Scale rank decreases or another positive-mass profile is created.
Profile splitting	Finite family of active or compound profiles and a remainder	Use mass decoupling, group same-frame profiles, and route the remainder by perturbative or new-core alternatives	A positive mass threshold bounds the number of active profiles.
Weak-to-strong failure	Nonzero compactness defect or residual concentration measure	Apply local compactness on a smaller cylinder or extract another concentration core	Failure becomes a new charged object rather than an unjustified strong-convergence assertion.
Pressure or gauge defect	Near pressure, harmonic tail, normalized gauge, and atomic/diffuse alternatives	Close locally by elliptic estimates, activate a velocity core, or route the surviving pressure defect to its own exhaustion	Pressure nonlocality is separated from velocity compactness and counted once.
Terminal-time escape	Endpoint concentration and no-escape quantity	Derive terminal no-escape from a local endpoint estimate or route to the complementary blow-up regime	Prevents normalized concentration from disappearing at the last time slice.

Representative locations make the routing index checkable without reproducing the companion arguments. Concentration, terminal no-escape, local compactness, and pressure gauges appear at nodes [5]–[10] of `proof_setup.tex`. Escaping-window and diffuse-defect compactifications appear at nodes [46]–[72] of `type_I_residual_closure.tex`. The profile, scale, and residual alternatives appear at nodes [47]–[52] of `type_II_regularity.tex`, while its pressure and gauge exhaustion is recorded at nodes [111]–[126]. These references identify the encoded routes; their local analytic evidence remains in the companion source.

This routing is constructive. Suppose a global compactness estimate needed by the argument is unavailable. Structural exhaustion assumes a normalized failure and applies the rows of [Table 7](#) exhaustively. When the required local lemmas hold, compact branches provide the local estimate. Noncompact branches either change to a frame in which the defect is visible, create a core that consumes critical mass, lower the active scale rank, or separate a pressure component governed by a local elliptic estimate. If the cited decoupling estimates supply a uniform positive charge and the scale order supplies a finite rank, these quantities bound the number of live successors. Once the local transitions, resource inequalities, and terminal contradictions are certified, [Theorem 3.5](#) excludes the normalized failure and propagates the resulting estimate to the root. This is the precise conditional sense in which the methodology constructs a global estimate through local mechanisms and exhaustive loss-of-compactness routes.

The companion developments are not independently validated here as a proof of global regularity, and this paper draws no Navier–Stokes regularity conclusion from them. Their role is to expose, at realistic scale, the state interfaces, resources, and closure routes that an LLM must maintain. Branch-preserving repair retains a reviewed continuation on the positive arm of a failed assertion, while structural exhaustion works on the defect arm using the same local-to-global calculus.

The source repository also contains the broader structural-exhaustion development from which the present PDE specialization is derived [\[DB26\]](#). The present paper isolates the human–AI protocol, the quantitative closure principles, and the PDE state variables needed to apply it.

7. DISCUSSION

7.1. What the method guarantees. Structural exhaustion has three levels of guarantee. Semantic soundness says that an exhaustive terminal graph proves the target on every admissible case. Quantitative closure propagates local constants and overlap bounds to a root estimate by [Theorem 3.2](#). First-failure compactness turns exclusion of every normalized defect into the existence of a uniform constant by [Theorem 3.3](#). When continuing defects consume a bounded resource or decrease a rank, [Theorem 3.4](#) also proves that the construction ends. [Theorem 3.5](#) combines these operations into the local-to-global estimate principle. These guarantees are mathematical consequences of the certified local transitions; the diagram records their composition.

If some leaves remain open, [Theorem 3.6](#) identifies the residual classes in which a counterexample may remain. If an asserted property fails, [Theorem 3.7](#) preserves the conditional mathematics below that property and routes its complement through the structural-exhaustion loop. These guarantees do not depend on a particular model or proof assistant.

The representation also supports useful automated checks: node identifiers can be compared with ledger rows; terminal shapes can be checked for outgoing edges; decision nodes can be required to list all alternatives; source links can be resolved; and state fields can be compared across transitions. Such checks detect structural omissions. They do not decide whether a cited PDE estimate is applicable.

For the Stokes graph included here, the repository audit verifies the contiguous range [\[1\]–\[9\]](#), nine solid nodes, one solid terminal, no closure proxies, the permitted number of outgoing edges for every node shape, and an audit-table row for every node. This is structural evidence only; the source ledger and the proofs of its cited lemmas provide the mathematical evidence.

7.2. Limitations. Structural exhaustion requires an exhaustive defect alphabet and a progress mechanism. A problem lies outside a proposed instantiation when a normalized failure produces a defect not represented by the state language, when a local transition has no proved estimate, or when continuing branches have no well-founded rank or bounded resource. These are precise mathematical design obligations. The method can replace a missing global theorem by composing local tools, but it cannot replace the proofs of the local dichotomies, compactness statements, capacity bounds, and rigidity results that make the composition valid.

A poor state choice can also create unnecessary branches or lose uniform constants. Interface-respecting descendants are essential: if a later proof uses an unstated feature of an extraction, it must be rewritten against an explicit topology, gauge, and quantitative interface. Aggregate transitions must record overlap and multiplicity; otherwise summing valid local estimates need not give a valid global estimate.

The Stokes example is linear and its local cutoff estimates and global kernel exclusion are explicit. The Navier–Stokes instances are substantially larger, but they do not constitute a controlled user study. Measuring whether the protocol improves correctness, review time, or AI contribution quality is future work. A useful evaluation would compare ordinary prose development with structural-exhaustion development on the same PDE problems and record omitted cases, repair cost, and expert review time.

8. CONCLUSION

Structural exhaustion gives human and AI collaborators a common unit of work: a local analytic state with a quantitative interface, a monotone resource, and an exhaustive continuation. Its central operation is constructive. Failure of a proposed global estimate is normalized, localized, and partitioned into defect mechanisms. Each mechanism supplies additional structure that can be estimated, charged, rescaled, compactified, or excluded by rigidity. Closing all routes produces the global estimate, while resource consumption makes the closure finite. Branch-preserving repair is a complementary benefit: a broken step preserves its conditional continuation and sends only its defect arm back through the same calculus.

The Stokes example shows why this matters even in a linear PDE: a local pressure split leaves a harmonic tail, while vorticity smoothing leaves a possible entire divergence–curl ambiguity. The local solenoidal estimate controls the raw velocity, and finite energy excludes the global ambiguity by an expanding-ball Caccioppoli argument. The equation then closes the pressure tail. The Navier–Stokes case study applies the same design to concentration, translation and scale escape, profile splitting, weak-to-strong defects, and pressure gauges. It shows how local analytic mechanisms can be assembled into global control without asserting that the companion branches constitute a validated regularity proof.

This is the role of structural exhaustion in AI-assisted PDE mathematics: the model maintains the large local state space and proposes typed failure routes; the proof derives its force from exhaustive local estimates, quantitative resource accounting, and terminal analytic contradictions. The methodology thereby occupies the space between an unsupported global guess and a fully formal proof script.

STATEMENT ON AI ASSISTANCE

Language models were used during manuscript editing and for consistency checks. The author selected the mathematical claims, verified the analytic arguments, and is responsible for the final text. No model output is treated as mathematical evidence merely by virtue of being generated.

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